Linear inequalities

1

a A dashed line **b** A solid line **c** A solid line **d** A dashed line

e A dashed line **f** A solid line **g** A solid line **h** A dashed line

2

a $x \geq 2$ **b** $y \leq 5$ **c** $y \geq -2x - 6$ **d** $y < 2x + 6$

e $y > -3x$ **f** $y \leq 2x + 5$

3 No

4 Yes

5

a **B** **b** **C** **c** **D**

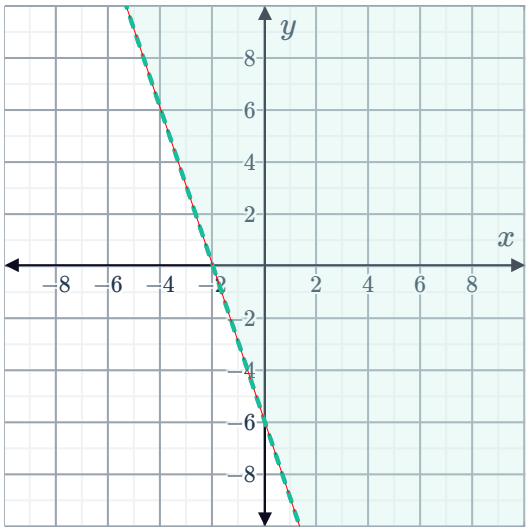
Graphing linear inequalities

6

1. Replace the inequality symbol with an equals sign and graph the linear equation.
2. Choose a test point from one of the half planes that is not on the line. Substitute its coordinates into the inequality.
3. If a true statement results, shade the half plane containing the test point, if false then shade the other half plane.

7

a $x = -2, y = -6$ **b** **C**

c 

d No

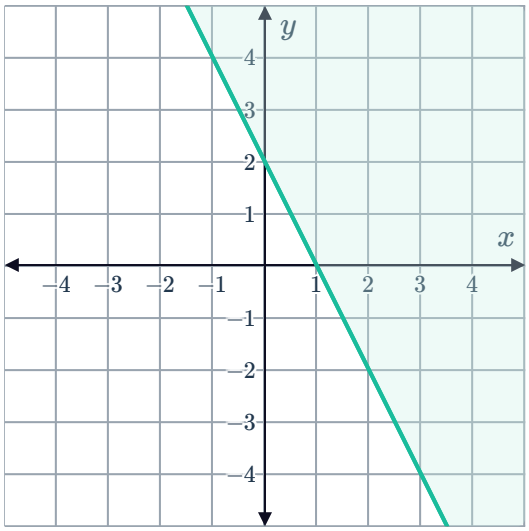
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8

a $x = 1, y = 2$

b **B**

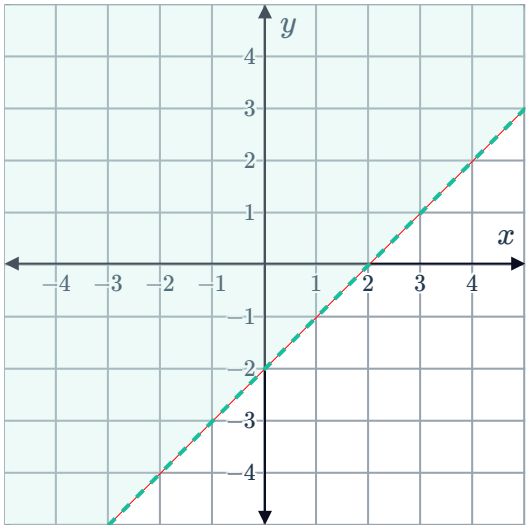
c



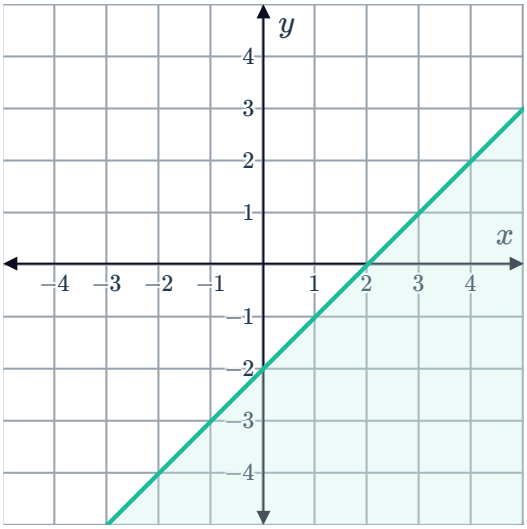
d Yes

9

a



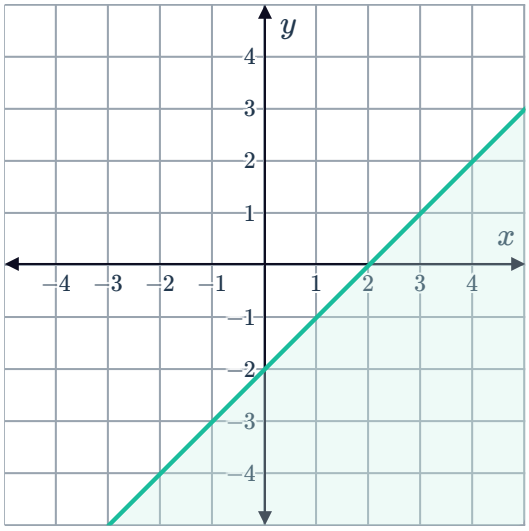
b



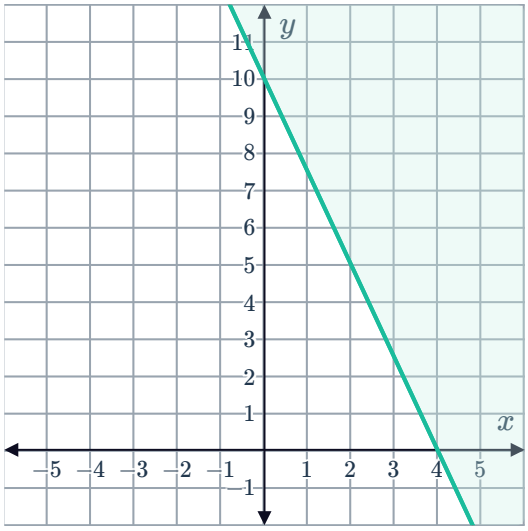
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10

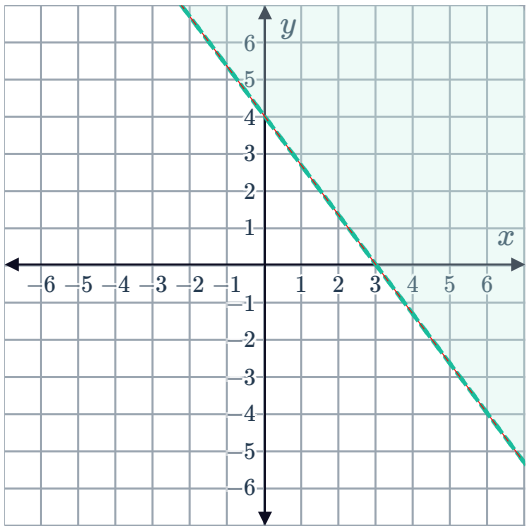
a



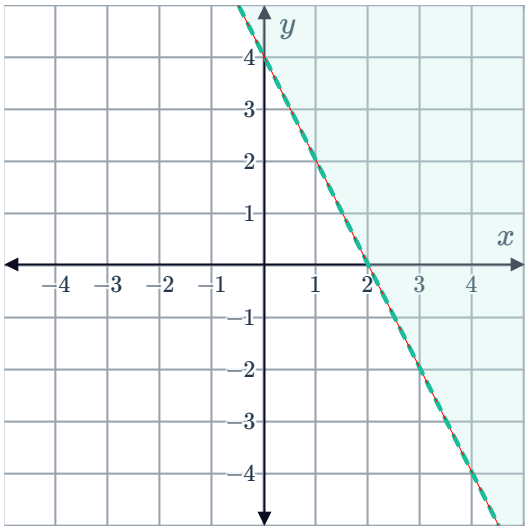
b



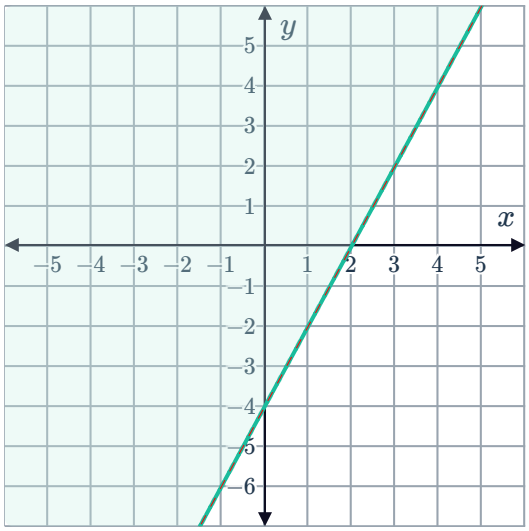
c



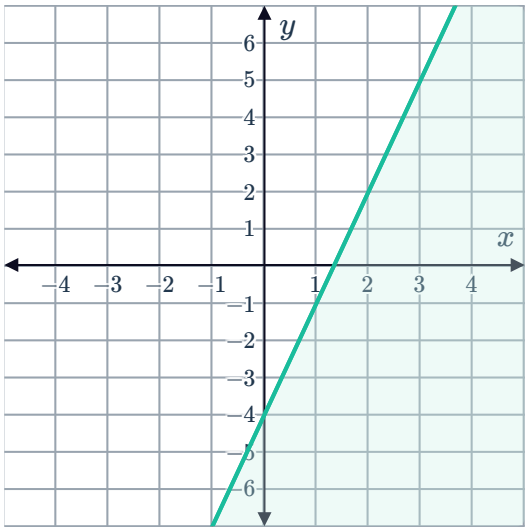
d



e

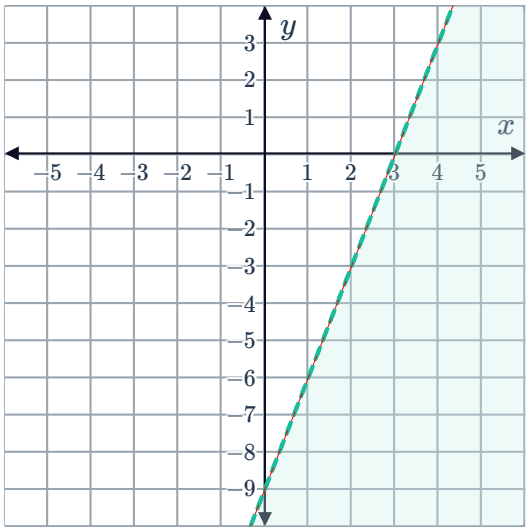


f

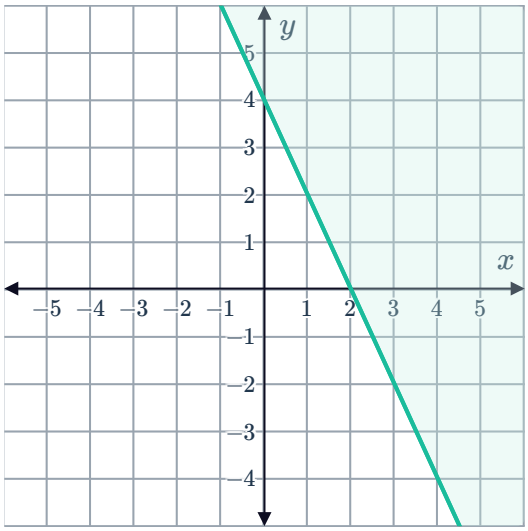


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

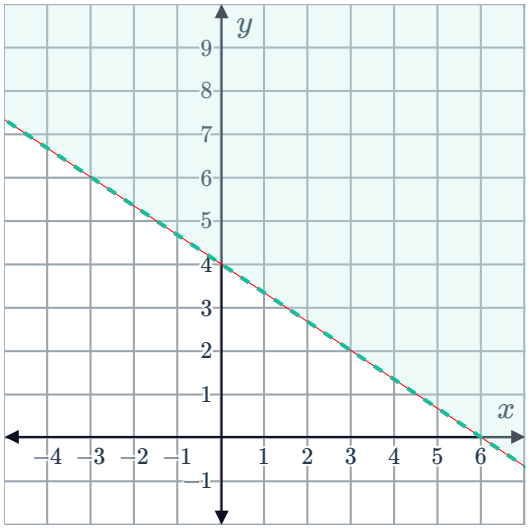
g



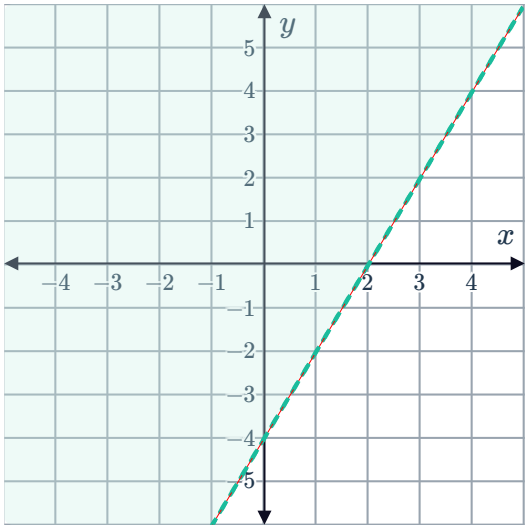
h



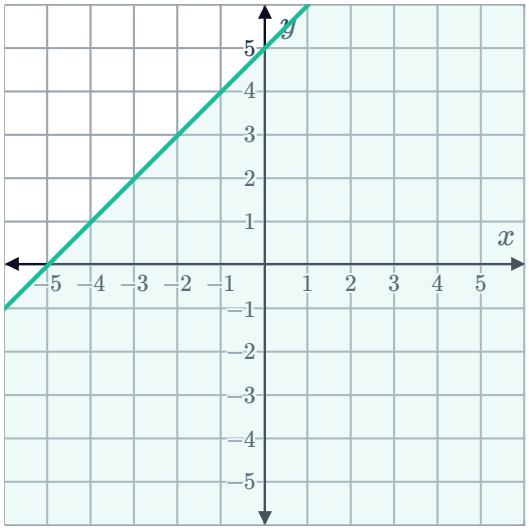
i



j



k



11

a $y \leq 4x - 5$

b $y \geq 3x - 6$

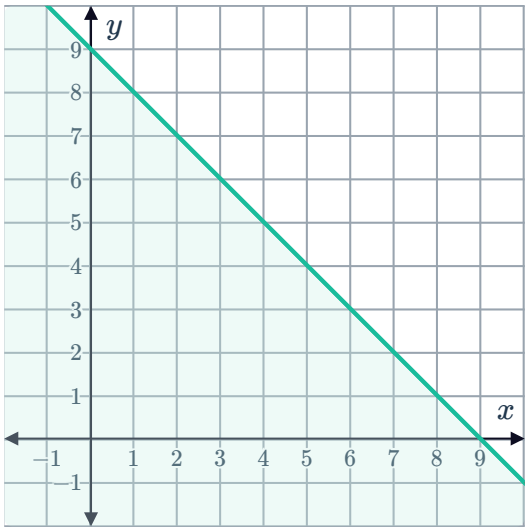
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Applications

12

a $y \leq 9 - x$

b



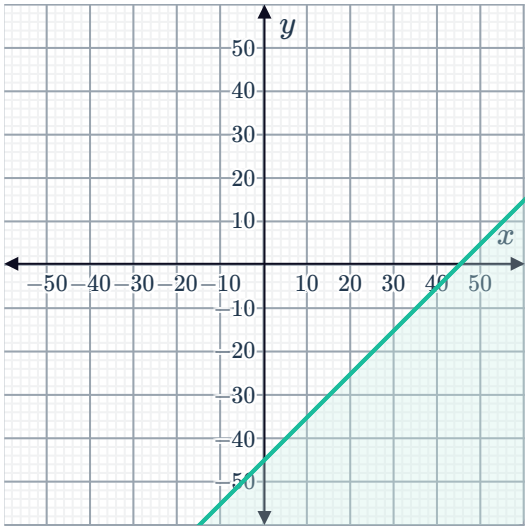
c

i No **ii** No **iii** Yes **iv** Yes

13

a $y \leq x - 45$

b



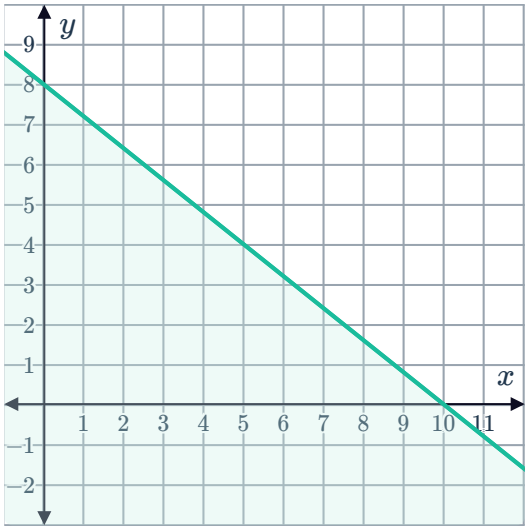
c 65 minutes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14

a $y \leq 8 - \frac{4}{5}x$

b



c

i No **ii** Yes **iii** No **iv** Yes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647